NeuroLens AI

Understand your signals. Notice patterns earlier.

AI for Human Health — UnivaBio Hackathon

1 Problem

People often experience health-related changes over time — difficulty concentrating, unusual forgetfulness, sleep changes, headaches, mood shifts — but these observations stay scattered across memory, notes, and daily life. People struggle to organize them and recognize meaningful patterns, and often don't know what information would be useful to discuss with a healthcare professional.

2 Solution

NeuroLens AI organizes everyday health observations into a structured timeline, analyzes patterns with a deterministic, explainable pattern engine, and uses AI to explain those patterns in simple, non-diagnostic language — turning scattered notes into better questions for healthcare professionals.

3 Key Features

- Health observation timeline — filterable by cognitive, sleep, mood, physical, daily functioning
- AI pattern analysis — frequency, trend, persistence, severity, impact, and correlation scoring
- Daily 60-second check-ins that update the timeline automatically
- Cognitive mini-exercises (memory, reaction, attention) — self-monitoring only, never clinical
- AI-generated insights — plain-language summary, context, clinician questions, monitoring suggestions
- Shareable health summary reports (PDF-ready, diagnosis-free, clinician-friendly)
- Privacy Center — export-my-data, delete-my-data, transparent AI processing

4 Technology

Frontend	React + TypeScript + Vite + Tailwind CSS + Recharts
Backend	FastAPI + Python (REST API, JWT auth)
Database	PostgreSQL (hosted) with SQLite fallback for local/demo
AI layer	Provider abstraction (OpenAI-compatible) + structured JSON responses
Engine	Deterministic, explainable pattern engine + safety-validation layer

5 Health Impact

NeuroLens helps users understand changes over time and prepare better questions to discuss with healthcare professionals — supporting earlier awareness and more productive clinical conversations.

6 Safety

NeuroLens is not a medical diagnostic tool. It never diagnoses disease, never prescribes medication, and never replaces professional medical care. Emergency-related input triggers an urgent-care notice, and every AI output is validated by an automated safety layer.

7 Innovation

Unlike a generic symptom chatbot, NeuroLens combines longitudinal health observations, deterministic pattern analysis, and AI-powered explanations — longitudinal pattern awareness, not one-shot symptom checking.

Live demo: https://frontend-itsaahsan-gmailcoms-projects.vercel.app · Source: https://github.com/itsaahsan/NeuroLens · Educational health insights — not a medical diagnosis.